Azure App Service, Azure AD and AWS IAM vs Azure AD RBAC

1. Deploy a Sample Application to an Azure App Service Deployment Slot

For this task, I worked with Azure App Service and created a separate staging environment for testing a sample application. The main idea was to test the application without making changes directly to the production environment.

Steps I followed:

1. I logged in to the **Azure Portal** using my account.
2. From the portal, I searched for **App Services** and opened the service.
3. I selected the option to create a new App Service.
4. I selected the subscription and resource group that I wanted to use.
5. I entered a name for my web application.
6. I selected the runtime that was suitable for my sample application.
7. I selected the required region and App Service plan.
8. After checking the details, I created the App Service.
9. Once the deployment was completed, I opened the newly created App Service.
10. I then looked for the **Deployment slots** option.
11. I created a new slot and named it **staging**.
12. The staging slot was created separately from the production environment.
13. I opened the staging slot and deployed my sample application there.
14. After the deployment finished, I opened the staging URL and checked whether the application was working.
15. I made sure that the application was running properly in the staging environment and that the production application was not affected.
16. I captured the required screenshots showing the App Service, staging slot, and application.

Result

I successfully created an Azure App Service and deployed the sample application to a staging slot. This helped me understand how applications can be tested separately before making changes to the production environment.

2. Register the Application in Azure AD

In this task, I registered the application with **Azure AD**, which is currently called **Microsoft Entra ID**. I used the application registration option to create an identity for the application.

Steps I followed:

1. I opened the Azure Portal and searched for **Microsoft Entra ID**.
2. I opened the service and selected **App registrations** from the menu.
3. I clicked on **New registration** to start registering my application.
4. I entered a suitable name for the application.
5. I selected the account type required for the application.
6. I provided the redirect URI when it was needed for the application.
7. After entering the required details, I clicked **Register**.
8. Once the registration was completed, I checked the **Application (client) ID** and **Directory (tenant) ID** generated by Azure.
9. I noted these details because they are useful when connecting the application with Microsoft Entra ID.
10. I checked the authentication settings and made the required configuration.
11. I verified that the application registration was available successfully in the App registrations section.
12. Finally, I took a screenshot of the registration details for my assignment.

Result

The application was successfully registered in Microsoft Entra ID. I also understood how the client ID and tenant ID are used to identify an application in Azure.

3. AWS IAM vs Azure AD RBAC Comparison

For the final part, I compared **AWS IAM** and **Azure RBAC** to understand how access and permissions are handled in the two cloud platforms.

Steps I followed:

1. I first went through the basic purpose of **AWS IAM** and **Azure RBAC**.
2. I looked at how users and groups are managed in AWS and Azure.
3. I checked how permissions are given to users and other identities.
4. I compared AWS IAM policies with the roles available through Azure RBAC.
5. I also looked at how access can be given to different cloud resources.
6. While comparing them, I noticed that both are mainly used to control who can access resources and what actions they are allowed to perform.
7. I noted the main differences in the way permissions and roles are defined in AWS and Azure.
8. Finally, I prepared a short comparison based on what I learned.

Result

Through this comparison, I got a basic understanding of how AWS IAM and Azure RBAC manage access to cloud resources. Although both are used for access control, they use different methods for defining and assigning permissions.